

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

df = pd.read_csv("marketing_sales_data.csv")

df.head()

      TV      Radio  Social Media Influencer      Sales
0   Low  3.518070      2.293790      Micro  55.261284
1   Low  7.756876      2.572287      Mega  67.574904
2   High 20.348988      1.227180      Micro 272.250108
3  Medium 20.108487      2.728374      Mega 195.102176
4   High 31.653200      7.776978      Nano 273.960377

print(df.shape)

df.info()

df.describe()

(572, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   TV              572 non-null   object
 1   Radio           572 non-null   float64
 2   Social Media    572 non-null   float64
 3   Influencer      572 non-null   object
 4   Sales           572 non-null   float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB

      Radio  Social Media      Sales
count  572.000000    572.000000  572.000000
mean    17.520616      3.333803  189.296908
std      9.290933      2.238378   89.871581
min      0.109106      0.000031   33.509810
25%     10.699556      1.585549  118.718722
50%     17.149517      3.150111  184.005362
75%     24.606396      4.730408  264.500118
max     42.271579     11.403625  357.788195

```

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df.isnull().sum()

TV          0
Radio       0
Social Media 0
Influencer  0
Sales       0
dtype: int64

print("Total Missing Values:", df.isnull().sum().sum())

Total Missing Values: 0

df_encoded = pd.get_dummies(
    df,
    columns=["TV", "Influencer"],
    drop_first=True
)

df_encoded = df_encoded.astype(float)

df_encoded.head()

```

|   | Radio     | Social Media | Sales      | TV_Low | TV_Medium | Influencer_Mega \ |
|---|-----------|--------------|------------|--------|-----------|-------------------|
| 0 | 3.518070  | 2.293790     | 55.261284  | 1.0    | 0.0       | 0.0               |
| 1 | 7.756876  | 2.572287     | 67.574904  | 1.0    | 0.0       | 1.0               |
| 2 | 20.348988 | 1.227180     | 272.250108 | 0.0    | 0.0       | 0.0               |
| 3 | 20.108487 | 2.728374     | 195.102176 | 0.0    | 1.0       | 1.0               |
| 4 | 31.653200 | 7.776978     | 273.960377 | 0.0    | 0.0       | 0.0               |

```

    Influencer_Micro  Influencer_Nano
0                1.0                0.0
1                0.0                0.0
2                1.0                0.0
3                0.0                0.0
4                0.0                1.0

corr_matrix = df_encoded.corr()

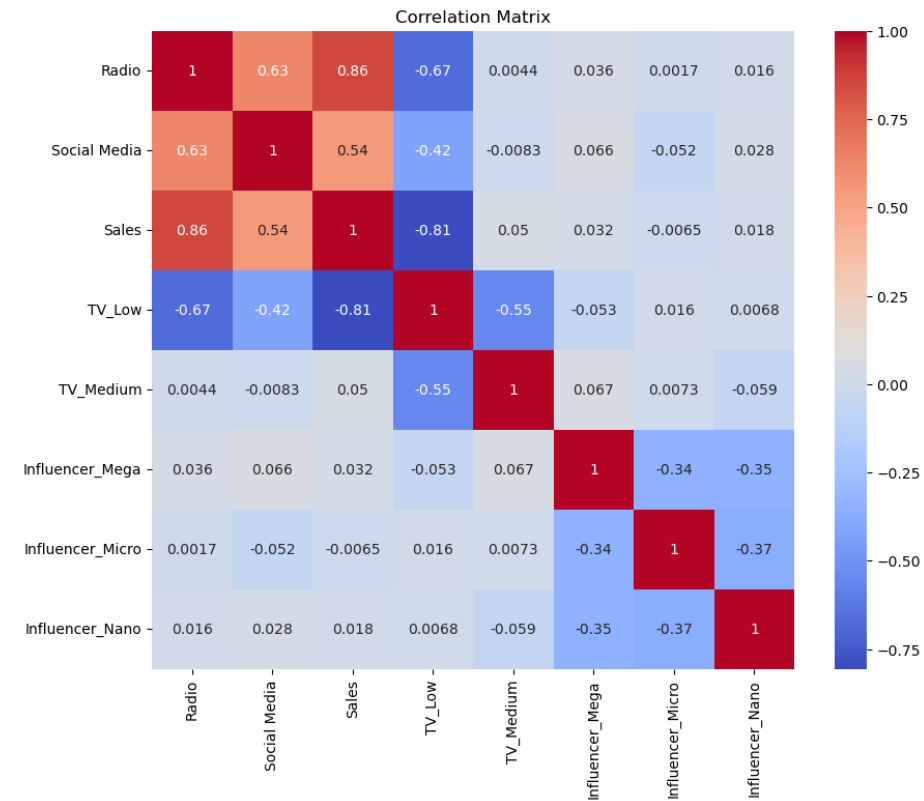
plt.figure(figsize=(10,8))

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

```

```
plt.show()
```



```
X_vif = df_encoded.drop("Sales", axis=1)
```

```
X_vif = sm.add_constant(X_vif)
```

```
vif = pd.DataFrame()
```

```
vif["Feature"] = X_vif.columns
```

```
vif["VIF"] = [
    variance_inflation_factor(
        X_vif.values,
        i
    )
    for i in range(X_vif.shape[1])
]
```

```
print(vif)
```

|   | Feature          | VIF       |
|---|------------------|-----------|
| 0 | const            | 31.649565 |
| 1 | Radio            | 3.479641  |
| 2 | Social Media     | 1.669098  |
| 3 | TV_Low           | 4.076384  |
| 4 | TV_Medium        | 2.233350  |
| 5 | Influencer_Mega  | 1.593889  |
| 6 | Influencer_Micro | 1.618434  |
| 7 | Influencer_Nano  | 1.627430  |

```
X = df_encoded.drop("Sales", axis=1)
```

```
y = df_encoded["Sales"]
```

```
X = sm.add_constant(X)
```

```
model = sm.OLS(y, X).fit()
```

```
print(model.summary())
```

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=====
                        OLS Regression Results
=====
Dep. Variable:          Sales      R-squared:                0.904
Model:                  OLS        Adj. R-squared:            0.903
Method:                 Least Squares   F-statistic:              760.4
Date:                  Wed, 17 Jun 2026   Prob (F-statistic):       1.82e-282
Time:                  11:34:20          Log-Likelihood:           -2713.4
No. Observations:      572              AIC:                     5443.
Df Residuals:          564              BIC:                     5478.
Df Model:               7
Covariance Type:       nonrobust
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|                  | coef      | std err | t       | P> t  | [0.025   | 0.975]   |
|------------------|-----------|---------|---------|-------|----------|----------|
| const            | 217.4784  | 6.584   | 33.031  | 0.000 | 204.546  | 230.411  |
| Radio            | 2.9735    | 0.235   | 12.644  | 0.000 | 2.512    | 3.435    |
| Social Media     | -0.1391   | 0.676   | -0.206  | 0.837 | -1.467   | 1.189    |
| TV_Low           | -154.5736 | 4.949   | -31.231 | 0.000 | -164.295 | -144.852 |
| TV_Medium        | -75.5947  | 3.647   | -20.726 | 0.000 | -82.759  | -68.431  |
| Influencer_Mega  | 2.4948    | 3.462   | 0.721   | 0.471 | -4.305   | 9.295    |
| Influencer_Micro | 2.9391    | 3.378   | 0.870   | 0.385 | -3.695   | 9.574    |
| Influencer_Nano  | 0.8015    | 3.346   | 0.240   | 0.811 | -5.770   | 7.373    |

```

=====
Omnibus:                 58.711   Durbin-Watson:              1.874
Prob(Omnibus):            0.000   Jarque-Bera (JB):            17.802
Skew:                     0.057   Prob(JB):                     0.000136
Kurtosis:                 2.143   Cond. No.                      147.
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Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
print("Adjusted R-Squared:", model.rsquared_adj)
Adjusted R-Squared: 0.9030011006551627
predictions = model.predict(X)

residuals = y - predictions
plt.figure(figsize=(8,5))

plt.scatter(
    predictions,
    residuals
)

plt.axhline(
    y=0,
    color="red"
)

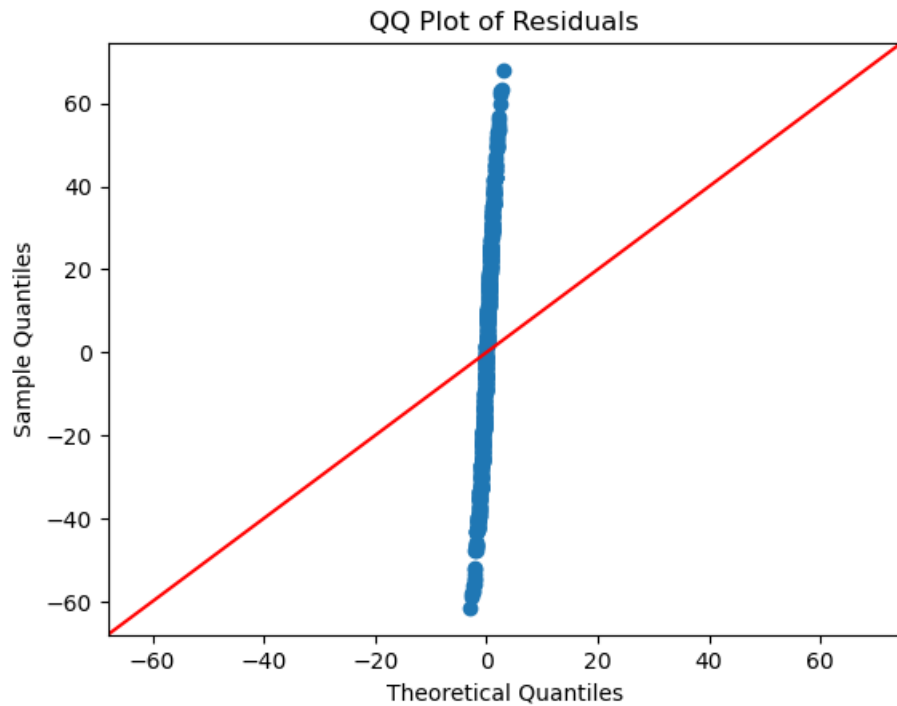
plt.xlabel("Predicted Sales")
plt.ylabel("Residuals")

plt.title("Residual Plot")

plt.show()
```



```
sm.qqplot(  
    residuals,  
    line="45"  
)  
  
plt.title("QQ Plot of Residuals")  
  
plt.show()
```



#### F-Statistic Interpretation

The model F-statistic is 760.4 with a p-value of  $1.82e-282$ . Since the p-value is significant, the model is statistically significant. This indicates that the predictors collectively provide

#### Business Recommendation

The regression model achieved an Adjusted R-squared of 0.903, indicating that approximately 90.3% of the variance in Sales is explained by the included marketing variables.

Radio advertising has a positive and statistically significant effect on Sales. TV advertising has a positive and statistically significant effect on Sales, with the High TV category outperforming Medium and Low TV campaigns.

Social Media and Influencer categories were not statistically significant predictors in this model. Therefore, the company should prioritize TV and Radio advertising channels while reassessing spending on Social Media and Influencer marketing.